SANA TILAK

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SUMMARY

Bachelor of Science degree candidate, with a focus in Chemistry and drug mechanics. I have experience with edveloping computational 2D QSAR benchmarking models, evaluating molecular fingerprints for machine learning driven drug discovery and analyzing chemical compounds. I look forward to continuing developing data-driven research, and helping scientists source their data much more efficiently. I hope to increase awareness for neglected diseases and contribute to their datasets.

EDUCATION

Borough of Manhattan Community College, New York City, NY — 2 years, 2026 - Present
A.S. in Science (Chemistry)

Azim Premji University, Bengaluru, Karnataka, India — 1 year (2025-26), transferred
BSc. (Honors) in Chemistry
Relevant Coursework: Introduction to Chemical Thinking, What is Biology, Chemicals in the Community, Mathematics and Computing
Awards: I was the chief founding member as well as the President of the inaugural American Chemical Society student chapter in this university.

Indian Institute of Technology Madras, Degree Program in Data Science and Applications — 4 years, 2025 - Present
BS in Data Science and Applications
Selected for Foundation in IIT Madras BS Degree Course (Through examinations), Cleared Foundation Level I, currently in Foundation Level II
Relevant Coursework: Programming in Python, Mathematics for Data Science II, Statistics for Data Science II

PROFESSIONAL EXPERIENCE

Personal Research Project

Rigorous 2D QSAR Benchmarking Study to Evaluate Molecular Fingerprints for Machine Learning-Driven Drug Discovery. Applied to all Neglected Tropical Diseases as underlined by the World Health Organization (WHO) — **2026**

- Evaluating how 5 distinct descriptor frameworks (Morgan/ECFP4, ECFP6, MACCS keys, RDKit topological paths, and Atom Pairs) affect predictive modeling accuracy across 8 ChEMBL neglected disease target datasets. Utilizing a scaffold-split approach and fixed Random Forest classifiers to isolate structural encoding efficiency via ROC-AUC, BEDROC, and MCC evaluation matrices verified with Wilcoxon signed-rank testing..

Personal Research Project, *Azim Premji University*

Identification of p-phenylenediamine in hair dye samples — **2025**

- Investigated the presence and concentration of p-phenylenediamine across various hair dye labels in an effort to detect which had the most carcinogenic effects on consumers (due to p-phenylenediamine and hydrogen peroxide reacting to form Bandrowski's base, a mutagenic compound).
- Utilized spectrophotometry for analysis.

Research Intern, *Claystation*, Glaze Chemistry/Testing Department — **2024**

- Interned with metallurgy experts at Claystation, a ceramic studio that produces their own glazes.
- Learned the basics of how ceramic glazes are made and their main components. Helped test new glaze products for the studio.
- Conducted a research case study on why different glazes produce different colors and compiled findings in a research paper.

Research Intern, *Indian Institute of Science*, Department of Chemistry — **2024**

- Interned at Professor DP Hari's Hari Group Labs. Shadowed his PhD students and assisted them with their research.
- Performed column and thin layer chromatography, took a look at how spectroscopy data is analyzed, and was able to see how Grignard reagents were used in experiments.
- Wrote a report on my learnings and experiences.

Attendee of Future Scientists Summer Camp, *Novo Nordisk*, Denmark **2023**

- Was one of 100 students selected from around the world to attend this program.
- Learned from various keynote speakers who spoke about diverse subjects ranging from biotechnology to sustainable plastic development.
- Dissected DNA from bacteria, took part in various experiments related to the body's functions, had an in-depth look into virtual reality, and more.

SKILLS

- *Laboratory Skills:* Perform spectrophotometry (spectroscopy), qualitative inorganic salt analysis, titrations, identify organic functional groups, experiments related to electrochemical cells, thin layer and column chromatography, pH-based tests, and tests that test for carbohydrates and proteins. Understands basic fundamental reactions that are used daily in the lab, and how to work with organic solvents.
- *Computational Skills:* Can work with basic python, understand algorithms and pseudocode. Basic data analysis, plotting, introductory QSAR concepts, handling molecular files (.sdf,. mol, etc.). Can design webpages and compile valuable information into bite-sized articles.

AWARDS

- *Academic Excellence Awards: 2022 - 2023*
- *School Topper for IGCSE - 2023*